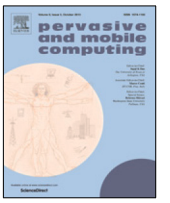
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MGA-Net+: Acceleration-based packaging work recognition using motif-guided attention networks



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ABSTRACT

This study presents a new method for recognizing complex human activities within the logistics domain, such as packaging operations, using acceleration data from a body-worn sensor. The recognition of packaging tasks using standard supervised machine learning is complex because the observed data vary considerably depending on the number of items to be packed, the size of the items, and other parameters. In this study, we focused on the characteristics and necessary key actions (motions) that occur during a specific operation. For instance, when the packaging tape is stretched while assembling the shipping boxes. To focus on these characteristic actions when recognizing data, we propose the use of an attention-based neural network. With our method, the attention-based neural network's training is guided such that its focus is on the motifs. In addition, this method was designed to accurately recognize short operations by leveraging data augmentation techniques. We tested our method on two logistics datasets and achieved a 3.9% improvement over the previous MGA-Net approach.
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1. Introduction

1.1. Background

E-shopping and delivery services have increased considerably in recent years, and these types of services now constitute most personal household purchases globally [1]. As a result, packaging logistics have become a central part of global and regional supply chains [1,2]. The creation of more efficient processes at these logistics centers makes it possible to significantly improve the entire supply chain. For example, because Amazon ships 2.5 billion packages every year in the U.S. alone, its costs related to shipping and handling processes are high. A considerable portion of these costs is generated inside the logistics centers, where the processes still rely primarily on manual activities by human workers, a trend that is expected to continue well into the foreseeable future [3–5], ensuring a greater level of flexibility and responsiveness to the ever-changing demands of customers and suppliers. Therefore, identifying and quantifying the actions conducted by logistics center workers is crucial for streamlining the overall process inside the center, for example, assessing and redesigning the workspace and reallocating resources.

In the past, human activity recognition (HAR) techniques have been used by researchers in the pervasive computing community to quantify manual tasks in the industrial domain, such as assembly works inside factories or packaging and

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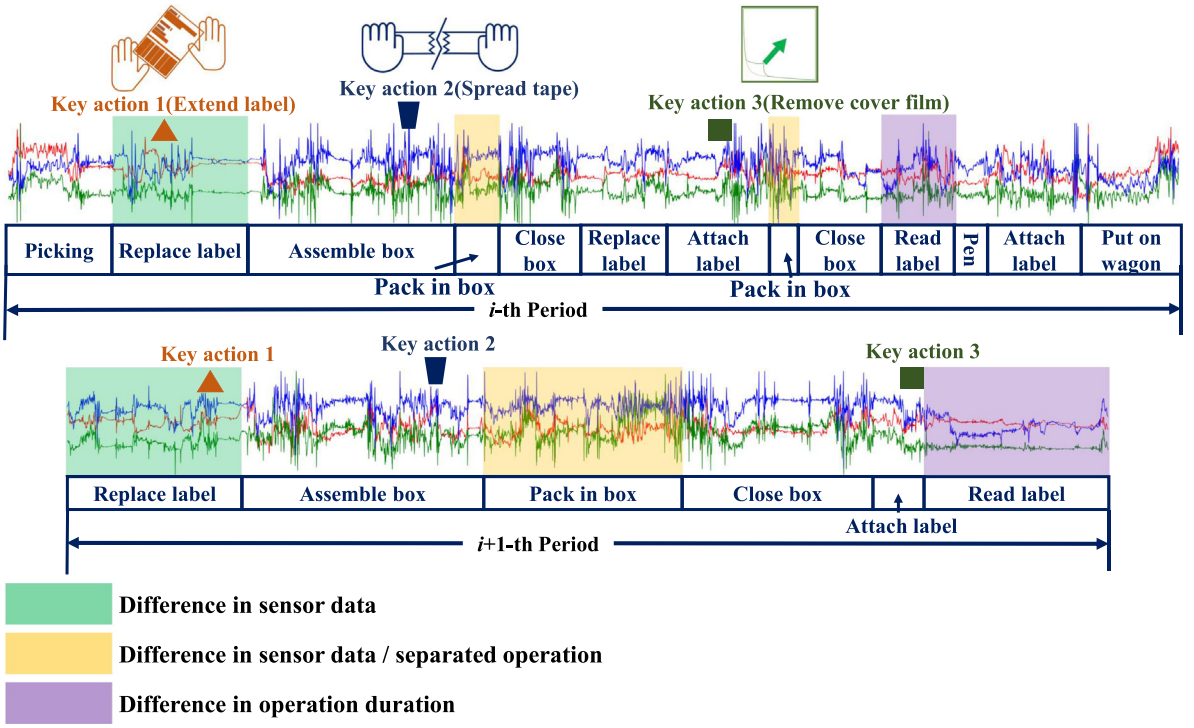


Fig. 1. Example sensor data regarding the work periods (ith and $i+1$-th periods). The red, green, and blue lines show the x-, y-, and z-axis data, respectively. The rectangles below show the ground truth labels of the operations. The highlighted sections are examples of differences between the same operation during different periods, which make the recognition task more difficult. The triangles, trapezoids, and squares show example key actions, which are characteristic actions found in specific operations (extending a label (Replace label), stretching tape (Assemble box), and removing glue protection (Attach label), respectively), regardless of the highlighted differences.

distribution tasks in logistics centers [6–9]. Inside a logistics center, when a customer's order is processed, an initial worker collects the items from the racks and delivers them to a second worker who packs them for delivery. During this packaging process, the worker must apply a repetitive sequence of operations, such as assembling the shipping box, placing the items inside along with cushioning material, and closing the box. This study focuses on identifying such operations using wearable sensor data and machine learning.

1.2. Problems

Fig. 1 shows packaging-related acceleration data collected from a worker's smartwatch at an actual logistics center. In this example, a typical series of operations is iterated twice, with each iteration (i.e., work period) comprising a sequence of 6–9 operations. The goal of this study is to predict an operation class label for each time slice.

The main challenges of this task are as follows. (i) **Difference in sensor data**: As shown in the example, the waveforms of an operation during different periods are dissimilar owing to the difference between the items to be packed. Each period has a number of items, each of which can vary in shape. (ii) **Difference in duration**: As shown in the example, the duration of an operation during different periods occasionally differs. For example, in the "Read label" operation during the ith period, the worker required less time to obtain the label information when using a laser scanner than during the $i+1$-th period. This is due to the close proximity of the scanner to the worker's hand after misplacing it during the previous period. (iii) **Complex data**: Several existing HAR studies assume simple waveforms, such as periodic waveforms, which are found while walking or running. By contrast, as shown in the example sensor data, the sensor data during packaging activities are highly complex because each operation involves multiple complex hand movements. (iv) **Amount of training data**: In addition to the aforementioned issues, we should address a problem related to supervised learning in the actual industrial domain. In the industrial domain, it is difficult to collect a large number of training data, such as with the ImageNet and YouTube-8M datasets. For these reasons, the training of deep models for industrial applications is a challenge.

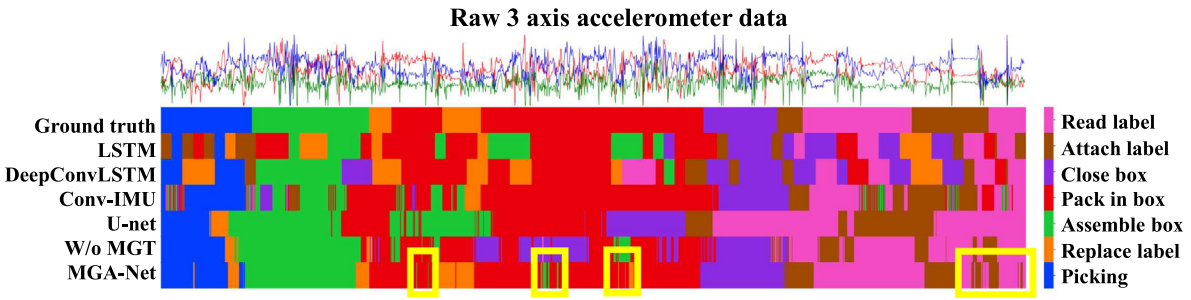


Fig. 2. Example of data period prediction for methods introduced in [9]. Highlighted squares show a high prediction variation and short label errors in multiple sections of the predicted period.

1.3. Approach

The recognition of packaging activities in a logistics center is a challenging task. However, our investigation of sensor data from actual logistics centers suggests that some characteristic actions are included in certain operations. For example, during an "assemble shipping box" operation, the worker stretches the packaging tape several times, an action that is not observed in other operations. In addition, during an "Attach address label" operation, the worker always removes the glue protection film from a label before attaching the label to a box. Fig. 1 shows examples of characteristic actions (key actions). We believe that these key actions can provide important clues for recognizing complex packaging tasks. We therefore proposed an attention-based neural network, called MGA-Net, that focuses on key actions for accurately recognizing operations [9]. Because the training of such complex attention models requires substantial training data, we employed motif-detection algorithms for the detection of sensor data motifs (characteristic sensor data segments corresponding to key actions), which can be useful for recognizing operations in advance. We then guided the training of the attention-based network such that it focuses on the detected motifs, that is, motif-guided training. With this method, we propose three types of motifs that are useful for recognizing packaging activities: unique, supportive, and boundary motifs. A unique motif is one that almost exclusively occurs during an operation of interest. A supportive motif is one that occurs before or after a unique motif of interest, with the time difference from the corresponding unique motif being consistent. The supportive motif helps to understand the temporal/sequential structures of key actions. A boundary motif corresponds to an action conducted around the start or end of an operation of interest. Identifying the start and end times of each operation is important for accurately recognizing sequential operations.

Despite this, important issues in time series segmentation remain unresolved, including (i) temporal smoothness and (ii) the data imbalance problem in under-represented classes. The proof of these issues is shown in Fig. 2. Here, we observe an example time series of estimates output by MGA-Net (and other comparative methods) when recognizing the sensor data of a testing period. As shown in the figure, the estimates by MGA-Net (and other comparative methods) have sporadic errors, particularly during long operations, as indicated by the yellow rectangles. Although MGA-Net relies on the CNN-based U-Net architecture [10], the architecture fails to capture the temporal smoothness of operations. To deal with this issue, we improve the MGA-Net architecture such that it can also capture a temporal regularity by introducing light-weight recurrent neural networks (RNN). In addition, the time series segmentation task suffers from a data imbalance problem. Because these methods assume a time series of sensor data for an entire work period as an input, the number of data points belonging to short-duration operations is inevitably small, resulting in a poor accuracy for such operations. On average, underrepresented operations have a weighted F1-score that is 13% lower than that of other operations. To address this issue, we leveraged data-augmentation techniques. Note that because the input of MGA-Net is time series sensor data of a work period, we cannot simply increase the number of data points belonging to short-duration operations. With our method, we generate a copy of the sensor data from a training work period and apply data augmentation techniques to sensor data segments corresponding to short-duration (under-represented) operations, enabling an increase in the variety of training data regarding such operations. The copied and augmented sensor data are then used as additional training data for the neural network.

1.4. Contributions

(i) We propose a new activity recognition method that guides the training of an attention-based network such that it pays attention to important sensor data motifs, that is, unique, supportive, and boundary motifs. (ii) We improve the original MGA-Net by introducing recurrent neural networks and data augmentation techniques. (iii) We investigate the effectiveness of the proposed method using sensor data collected at an actual logistics center.

Activity	right arm	left arm	right arm reach forw.	left arm reach forw.	right arm reach back	left arm reach back	right arm stretched out	left arm stretched out	arms push	arms pull	upper body move forw.	upper body move back	legs walk	legs stand	body pose box	body pose article	body pose cart
Walking (none)	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
Walking (box, both-handed)	1	1	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0
Standing (none)	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
Standing (box, both-handed)	1	1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
Reaching forward (none, both-handed)	1	1	1	1	0	0	0	0	0	0	1	0	0	1	0	0	0
Lifting (box, both-handed)	1	1	0	0	1	1	0	0	0	0	0	1	0	1	1	0	0
Putting down (box, both-handed)	1	1	1	1	0	0	0	0	0	0	1	0	0	1	1	0	0
Straightening up (none)	0	0	0	0	1	1	0	0	0	0	0	1	0	1	0	0	0

Fig. 3. Example of an expert-given attribute representation for the activities in the LARa dataset.

2. Related work

2.1. Human-activity recognition in industrial settings

Because of the increasing interest in smart manufacturing and Industry 4.0 [11–13], methods for identifying and improving factory work using wearable sensor technologies have been extensively studied [6–8,14–16]. For example, Koskimäki et al. collected data from wrist-worn accelerometers to ensure that all required actions are conducted [17]. They identified operations such as screwing and hammering using a *k*-nearest neighbor search. Ward et al. recognized individual operations using Hidden Markov Models (HMMs) and a linear discriminative classifier from inertial acceleration and audio data collected during woodworking activities [18]. Stiefmeier et al. used acceleration data acquired during manual work, such as maintenance work and bicycle repair, and employed an HMM or template matching to recognize the data [19,20].

Rueda et al. employed a parallel approach based on a Convolutional Neural Network (CNN) for multiple inertial measurement units (IMUs) [7]. Their method involves preparing a block composed of convolutional and max-pooling layers for each IMU, and then merging all parallel paths for the IMUs on a single fully connected layer. This method was evaluated using sensor data collected on a simulated logistics scenario.

Reining et al. used CNNs to identify the attribute representations of activities conducted during the order-picking process [8]. They employed CNNs to process motion capture data and recognize specific motions that are considered characteristic features of multiple activities in an order-picking system.

However, as can be seen in Fig. 3 this approach requires a precise association between the attributes and activity classes created by an expert. Tao et al. employed CNNs to recognize six different standard consecutively applied assembly operations. They converted multichannel IMU data and electromyographic data into images and trained the CNNs on the images [6]. By contrast, we enhanced the training of attention-based neural networks using motif-guided training. To address the scarcity of the training data, Xia et al. developed a motif-based particle filter for recognizing factory tasks in an unsupervised manner in a real work environment [15,21]. However, the aforementioned studies assume that workers sequentially conduct predefined operations in a predefined order within the expected duration of each operation specified in an instructional document, or they require the researchers in charge of defining the attributes to be familiar/experts in the associated industrial context.

2.2. Motif detection for human-activity recognition

Several studies have explored motif detection algorithms for activity and gesture recognition [22–24]. Minnen et al. discovered motif seeds using a minimum description length (MDL), and then refined the motif seeds by splitting, merging, and extending the motifs [22]. Maekawa et al. measured the duration of each work period on a production line in an unsupervised manner by identifying a motif (corresponding to an atomic action) that appears only once during each work period [24]. Dissanayake et al. predicted the location classes of a room, such as a kitchen or restroom, where a subject is located by discovering location-specific sensor data motifs through the calculation of a score that represents the "location specificity" of each motif [25]. In our previous study [9], we first extracted motifs corresponding to the actions based on these algorithms. Subsequently, we used the extracted motifs to efficiently train attention-based neural networks. In this study, based on prior research, we propose a new method for addressing the two issues inherent to time series segmentation mentioned in the Introduction. Table 1 compares the proposed method (MGA-Net+) with the prior version (MGA-Net) in terms of several aspects. Other methods investigated in the evaluation section are also listed in the table.

Table 1
Comparison of different methods. We compare the attributes between previous methods developed for human activity recognition and our approach. See the evaluation section for further details regarding these methods. Note: The number of trainable parameters corresponds to an input size consistent with the longest existing period from the LOGI dataset, which is also introduced in the evaluation section.

Method	Network backbone	# trainable parameters	Use of attention	Use of motif	Use of data augmentation
ConvLSTM	Convolutional*RNN	26,070	No	No	No
U-Net	U-Net	32,248	No	No	No
W/o MGT	U-Net	106,401	Yes	No	No
MGA-Net	U-Net	106,401	Yes	Yes	No
MGA-Net+	U-Net+RNN	115,092	Yes	Yes	Yes

2.3. Gated recurrent units for human activity recognition

Modern supervised learning approaches using convolutional and/or recurrent neural networks have been increasingly utilized and have demonstrated improvements in the classification accuracy over conventional models [26–28]. Studies, such as those conducted by Yang et al. integrated Conv1D and Gated Recurrent Units (GRUs) to extract spatial and temporal features, respectively [26]. These studies show that CNN-GRU is effective in human activity recognition under traditional circumstances. Similarly, for the case of multi-sensor data streams, Nafea et al. proposed the use of hierarchical multi-resolution convolutional neural networks in combination with gated recurrent units to classify more complex actions, such as healthcare surveillance for patient monitoring in care facilities [28]. Such studies have proven that GRUs can extract different types of features than those extracted using standard convolutional networks. However, no related studies have employed this technique in industrial settings or for the classification of complex actions. As shown in Table 1, in this study, we enhanced the original MGA-Net structure presented in [9] with two separate bi-directional GRUs improving the ability of the method to smooth longer activity classifications while only slightly increasing the number of parameters.

2.4. Data augmentation for timeseries classification

In traditional deep learning models, the use of large numbers of data offers a reliable base for training and improves the classification. However, in the field of time series recognition, many datasets are often extremely small, as is the case with our available real-life data. Studies presented by Iwana et al. and Le et al. demonstrated the benefits of applying different types of data augmentation on time series data for HAR on convolutional networks [29,30]. The seven types of augmentation presented by Iwana et al. proved to be significant in classifying normal human actions while preventing an overfitting of the network [29]. Another survey study conducted by Wen et al. showed that simple data augmentation approaches (e.g., time and frequency) have a larger impact when several handpicked augmentation types are combined for tasks involving time series classification [31]. When dealing with class-imbalanced datasets, one classical approach, as presented by Fernandez et al. is to oversample the minority class using the synthetic minority oversampling technique to artificially mitigate the imbalance [32]. Based on such studies, to deal with our imbalanced dataset, we generate new data for the minority classes while combining several traditional data augmentation techniques to boost the number of overall training data and limit the impact of class imbalance. The use of this selective data augmentation is the second improvement over MGA-Net and other previous methods applied to packaging task recognition, as shown in Table 1.

3. Methodology

3.1. Preliminaries

This study assumed that workers wear body-mounted accelerometers. Each worker sequentially processes orders from customers. The work period for processing an order includes one iteration of the overall operations, as shown in Fig. 1. We assumed that training data consisting of labeled sensor data segments were available, with each labeled segment corresponding to the sensor data of one training (work) period. Each label attached to a segment specifies the start and end times of an operation included during the period and a class label of the operation, as shown in Fig. 1.

3.2. Overview

The proposed method involves training and testing phases. During the training phase, as depicted in Fig. 4, we first preprocess the sensor data of the training periods (standardization) and then find unique, supportive, and boundary motifs from the training data. For each motif, we calculate the motif-occurrence sequence from the sensor data for each training period. A motif-occurrence sequence is a time series that specifies when a motif of interest occurs. We then apply data augmentation to increase the sensor data variation in short-duration operations. We train our motif-guided attention

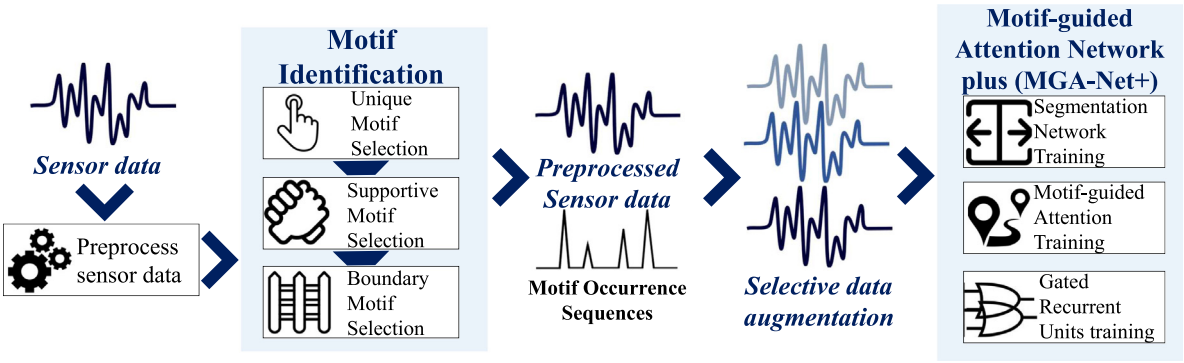


Fig. 4. Overview of the training phase of the proposed method.

network plus (MGA-Net+) on labeled sensor data and motif-occurrence sequences. During the test phase, we preprocess a sensor data segment of the test period and then feed the preprocessed segment into the trained MGA-Net+. MGA-Net+ outputs a class label for each data point in the input segment.

3.3. Motif identification

Herein, we symbolize the input time series by assigning a symbol (letter) to each small section in the time series based on the average magnitude of the data points. We then find useful motifs from the sequence of symbols created. We do this by extracting candidate motifs with l_m symbols from the created sequence corresponding to the initial working period in the training data. To extract the candidate motifs, we use sliding time windows with a stride of 1 (symbol) and a window size of l_m (symbols). Note that candidate motifs for each operation are extracted from the segment of the first training period that corresponds to that operation. We then compute an occurrence sequence for each motif candidate by checking their similarity to the sequence of symbols for each training period. Thereafter, we calculate the scores of each candidate to identify unique, supportive, and boundary motifs among them. Finally, we select a unique, a supportive, and a boundary motif for each operation. Fig. 5 illustrates the procedures for symbolization, extraction of candidate motifs, and the calculation of motif-occurrence sequences.

3.3.1. Preprocessing

First, we employ a principal component analysis (PCA) [33] to reduce the dimensionality of the sensor data. Because the three-axis time series data collected from a worker were used in our experiment (acceleration data from a sensor attached to the dominant hand), we take the first principal component and use the resulting one-axis time series for motif identification. The time series created is shown in (a) in Fig. 5.

To efficiently find the motifs, we preprocess the created one-dimensional time series. We first use a piecewise aggregate approximation (PAA) [34] to reduce the number of data points within the time series. In PAA, data points in a data segment within a time window with a length of δ are averaged, and this average is then used as a representative value of the window. This process can be seen in the transition from (a) to (b) in Fig. 5.

We then symbolize the downsampled data, as shown in Fig. 5 (b). In brief, we convert each downsampled numerical value into a symbol based on the value range associated with each symbol [35]. For example, a time series is symbolized as *aabcddbaa*, where the same letter belongs to the same value range.

We then extract motif candidates with a length of l_m for each operation. First, we obtain a symbol sequence corresponding to the operation of interest during the first training period. Next, we use a sliding window with a length of l_m and a stride of one symbol to extract motif candidates with a length of l_m from the symbol sequence, as shown in Fig. 5 (c).

3.3.2. Calculating occurrence sequence of motif

From the calculated sequence of symbols for each training period, we compute the occurrence sequence of each extracted candidate motif using a sliding time window, as shown in the transitions from (d), (e), and (f) in Fig. 5. The length of the window is l_m, and the window moves along a sequence of symbols with a stride of 1, as shown in (a). To obtain a series of similarity values, we calculate the similarity between the extracted candidate motif and the series of symbols within each time window. When calculating the similarity, we leverage the Levenshtein distance metric [36], which is widely used to compare two symbol sequences. We can see the result of such calculation in (e).

We then process the similarity sequence to transform it into an occurrence sequence. Given that we require the positions where a key action (motif) is conducted, we convert all similarity values below the threshold th_s to zero.

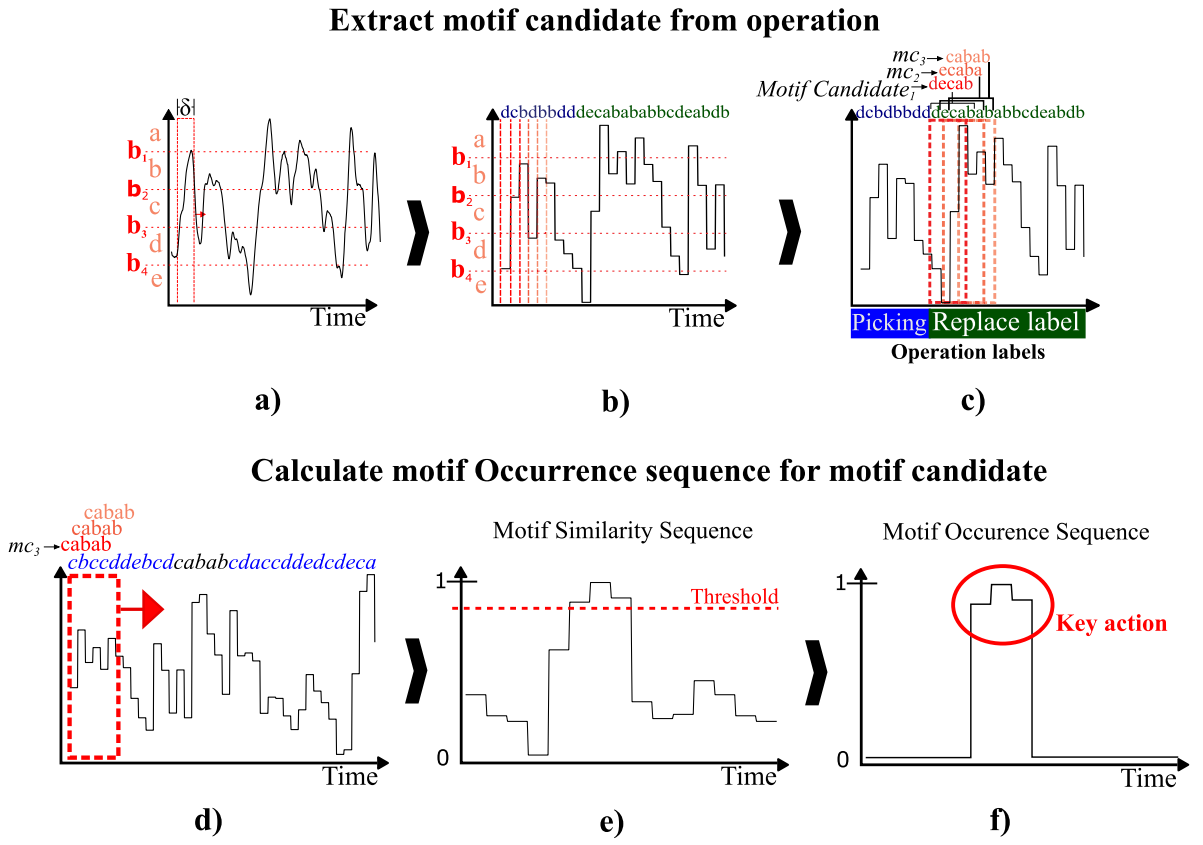


Fig. 5. Overview of motif extraction and motif occurrence sequence calculation. Top: three steps for candidate extraction, i.e., (a) piecewise aggregate approximation, (b) window symbolization, and (c) sliding window motif extraction. Bottom: example of steps used to obtain a candidate occurrence sequence for Motif Candidate 3 (mc_3), i.e., (d) sliding window similarity calculation, (e) focus on high similarity sections, and (f) a segment of the motif occurrence sequence when finding a key action (spread tape).

As a result, we obtain an occurrence sequence, which is a numeric time series exhibiting high similarity values at the occurrence point of the motif of interest, as shown in (f).

After calculating an occurrence sequence for each motif candidate of each operation for each training period, we discard motifs that do not frequently occur in the operation of interest. Finally, for each target operation, we select the 10 most frequent motif candidates for each occurrence of the operation of interest during all subsequent training periods. We use this shortlist to select the three defining motifs (unique, supportive, and boundary) for each operation.

3.3.3. Identifying unique motif

A unique motif that usually appears during a specific operation is useful for identifying that operation. Moreover, a unique motif should not be observed during any other operations. Therefore, we calculate a uniqueness score for each candidate motif belonging to operation O_j using the motif-occurrence sequences of the candidate motif extracted from the training data:

$$\mathbf{U}_{score} = \frac{1}{N_t} \sum_{i=1}^{N_t} \frac{\mathbf{o}_i \mathbf{B}_i^{\mathrm{T}}}{T^i} \tag{1}$$

where N_t is the number of training periods, T^i is the length of the ith period, $\mathbf{o}_i$ is the occurrence time series of the ith period for the candidate motif, and $\mathbf{B}_i$ is a time series whose element value at time t is 1 when the operation label matches that of the motif candidate, and is -1 otherwise.

$$\mathbf{B}_i^t = \begin{cases} 1, & \text{Operation at time } t \text{ is } O_j \\ -1, & \text{otherwise} \end{cases} \tag{2}$$

When the motif of interest occurs during the operation of interest, $\mathbf{o}_i \mathbf{B}_i^{\mathrm{T}}$ has a large positive value. By contrast, when the motif of interest frequently occurs in operations other than the operation of interest, $\mathbf{o}_i \mathbf{B}_i^{\mathrm{T}}$ takes a large negative

value because the value of $\mathbf{B}_i^t$ is negative when the operation of interest does not occur. We consider the candidate motif with the best score to be a unique operation motif. Note that, when identifying a unique motif, we vary the value of l_m, according to the test parameters in Table 4, to generate motif candidates (and their occurrence sequences) and select the candidate with the best score.

3.3.4. Identifying supportive motif

Although an operation consists of a sequence of actions, a unique motif can locate only a single type of action. It is therefore difficult to model the structure of an operation using only a unique motif. To efficiently recognize an operation, it is important to capture the sequential structure of the actions. We employ a supportive motif for the unique motif of operation O_j that occurs before or after the unique motif, where the time difference between the unique motif and supportive motif is consistent during each period. To find the supportive motif among the candidate motifs of operation O_j, we calculate a score for each candidate motif as follows: Note that this calculation is conducted only for candidates whose time difference from the unique motif is longer than t_{su} during the first training period.

$$S_{score} = \frac{|\mathbf{T}_{\boldsymbol{\Delta}}|}{N_t} \sqrt{\frac{\sum_i (\mathbf{T}_{\boldsymbol{\Delta}i} - \overline{\mathbf{T}_{\boldsymbol{\Delta}}})^2}{|\mathbf{T}_{\boldsymbol{\Delta}}|}} \tag{3}$$

where $\mathbf{T}_{\boldsymbol{\Delta}}$ is a vector representing the time difference between the occurrence of a unique motif and the nearest occurrence of a candidate motif during the training period. Here, $\mathbf{T}_{\boldsymbol{\Delta}i}$ denotes the ith element of $\mathbf{T}_{\boldsymbol{\Delta}}$. The differences are calculated by subtracting the timestamp of the candidate-motif occurrence from that of the nearest unique-motif occurrence within a period. By calculating $\sum_i (\mathbf{T}_{\boldsymbol{\Delta}i} - \overline{\mathbf{T}_{\boldsymbol{\Delta}}})^2$, we can measure the stability of the temporal difference between the occurrences of unique and candidate motifs. We select the candidate motif with the best score as the supportive motif for the operation.

3.3.5. Identifying boundary motif

Unlike the unique and supportive motifs, a boundary motif appears at the boundary between two consecutive operations. Because we employ a segmentation neural network to find the start and end times of each operation, it is important to know where the boundary between operations lies. A boundary motif for operation O_j is selected from among the candidate motifs of operation O_j that are not selected as a unique or supportive motif. We calculated a score for each candidate motif based on the temporal difference between the occurrence of the candidate motif and the end of the designated operation as follows:

$$B_{score} = \frac{|\mathbf{B}_{\boldsymbol{\Delta}}|}{\overline{\mathbf{B}_{\boldsymbol{\Delta}}} N_t} \sqrt{\frac{\sum_i (\mathbf{B}_{\boldsymbol{\Delta}i} - \overline{\mathbf{B}_{\boldsymbol{\Delta}}})^2}{|\mathbf{B}_{\boldsymbol{\Delta}}|}}, \tag{4}$$

where $\mathbf{B}_{\boldsymbol{\Delta}i}$ is the temporal distance between the ending time of operation O_j during the ith period and the nearest occurrence of the candidate motif. By calculating $\sum_i (\mathbf{B}_{\boldsymbol{\Delta}i} - \overline{\mathbf{B}_{\boldsymbol{\Delta}}})^2$, we can measure the stability of the occurrences of the candidate motif. In addition, $\overline{\mathbf{B}_{\boldsymbol{\Delta}}}$ in the denominator increases the score of a candidate that appears close to the boundaries. The candidate motif with the best score is selected as the boundary motif for the operation.

3.4. Sensor data augmentation

During this process, we generate copies of the training sensor data for each period to increase the variations in the sensor data. We can see an example training period in (a) of Fig. 6. Specifically, because the number of data points belonging to short-duration operations (i.e., under-represented classes) is limited, resulting in a poor recognition performance of these operations, we increase the variation in the sensor data of these short-duration operations. As can be seen in Fig. 6 (a), for each training period, we search for classes that are defined as under-represented and select them for data augmentation. In the case of example (b), "Picking" is the under-represented operation. We first generate N_{DA} copies of the sensor data for each training period ($N_{DA} = 7$ in our implementation). We then apply data augmentation techniques on each of the segments corresponding to short-duration operations in each copy while leaving the rest of the period data as-is. An example can be seen in 6 (d). For each segment, we randomly select a data augmentation operation (i.e., jittering, mirroring, or scaling, as shown in [29]), and apply the operation with randomly selected parameters. For each period of augmented data, we vary the augmentation parameters, such as the number of sensor axes to mirror, or the magnitude of scaling, regardless of whether there are one or more segments (classes) to be augmented.

The generated copies of the training data are used as additional training data for MGA-Net+, that is, $N_{DA} + 1$ versions of the same training period are used. Note that when we train MGA-Net+ on a copy of the sensor data, because we want MGA-Net+ to focus on the real characteristic motifs, despite being altered by the data augmentation techniques, we use the motif occurrence sequences generated from the original sensor data (see the next subsection on MGA-Net+ training for further details).

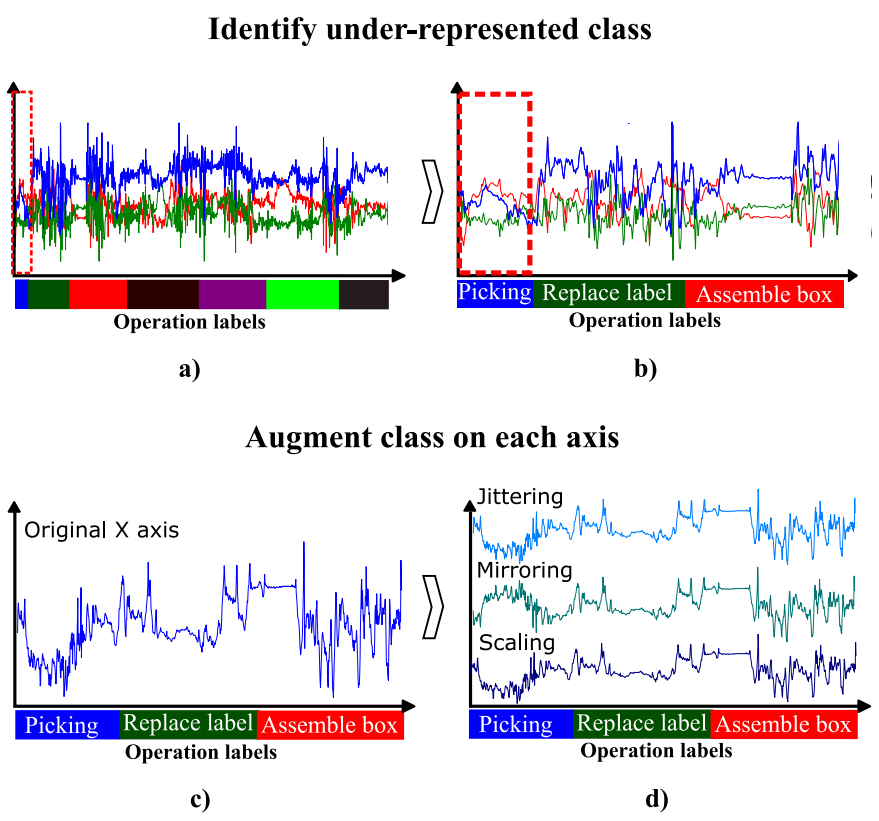


Fig. 6. Overview of selective data augmentation for short-duration (under-represented) operation classes. (a) An entire training period and the "Picking" operation being identified as under-represented. (b) The data segment corresponding to the "Picking" selected for augmentation. (c) The *x*-axis of the accelerometer time series before data augmentation. (d) Three copies of an augmented period after different operations were conducted on the "Picking" data.

3.5. Motif-guided attention network plus (MGA-Net+)

The input of our motif-guided attention network plus (MGA-Net+) is a three-dimensional time series of preprocessed sensor data. When we train MGA-Net+, the occurrence sequences of unique, supportive, and boundary motifs are also used to train the attention heads in MGA-Net+. The output of MGA-Net+ is a series of estimates (class labels) for each data point.

3.5.1. Network structure

Fig. 7 shows the structure of MGA-Net+, which is based on the U-Net topology for the segmentation of time series data [10]. The network consists of three encoding blocks and three decoding blocks. Each encoding block consists of two one-dimensional convolution layers and a max-pooling layer. Each decoding block consists of two one-dimensional convolution layers and one upsampling layer. Another part of the original U-Net topology is the concatenation layer at the beginning of each decoder block, where the output from the second convolutional layer of the corresponding encoding block is concatenated with that of the upsampling layer. This network structure enables the use of a complete work period of data as a single input. It also permits us to output an estimate (class prediction) for each data point in the input time series. As shown in Fig. 7, we include a multihead attention mechanism at the end of the first encoding block, i.e., f attention heads, to identify unique, supportive, and boundary motifs for each operation. This attention mechanism consists of one attention head dedicated to identifying each specific motif of an operation (i.e., three heads for each operation), with each attention head analyzing the input time series independently to focus more on identifying the assigned motif. We introduce this attention mechanism into the first encoding block to focus even on small actions that would be lost in the deeper layers. The attention layer is composed of $3 \times K$ attention heads, where K denotes the number of operation classes. Each attention head individually calculates the time series of the attention according to Eq. (5). The length of the time series of attention is identical to that of the output of the first encoding block. The time series of

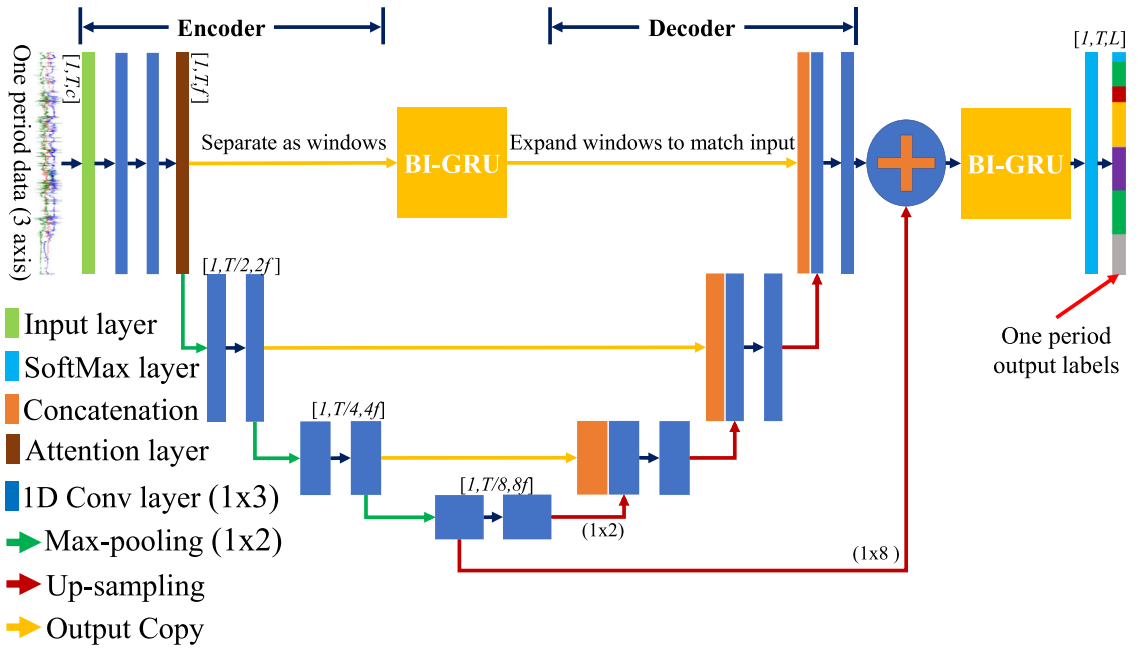


Fig. 7. MGA-Net+ structure. The model input is a time series with one period of raw accelerometer data, and the output is a vector with the operation label for each data point in the input time series (i.e., dense labeling). The filter sizes for each type of layer are shown in the parenthesis. the layer output dimensions are shown in the brackets, where f is the number of nodes (kernels) in the convolution layer in the first encoding block, T is the input length, c is the number of input channels ($c = 3$), and L is the number of class labels.

attention is calculated as follows:

$$\mathbf{a} = \text{softmax}(\tanh(WZ^{\mathrm{T}})), \tag{5}$$

where W represents the trainable parameters of the weight of the attention head, and $Z \in \mathbb{R}^{T \times f}$ is the output of the last convolutional layer in the first encoding block. In addition, T is the length of the input time series, and f is the number of nodes in the convolution layer in the first encoding block. The attention value in $\mathbf{a}$ at time t indicates the importance (i.e., attention) of the data point at t. To highlight important segments (corresponding to motifs) in the output of the first encoding block, the attention time series is multiplied by the output of the last convolution layer in the first encoding block. Note that different attention heads have different trainable weight parameters (i.e., W). To capture the temporal regularity of the operations, MGA-Net+ includes two bi-directional GRU blocks [27], as shown in Fig. 7. The first Bi-GRU block is introduced between the first layer of the encoder and the last layer of the decoder. This is introduced to feed long-term temporal information into the last layer of the decoder, which is responsible for outputting the final results. The second Bi-GRU block is introduced between the last layer of the decoder and the output layer. The inputs of this Bi-GRU block are the outputs of the final layer of the encoder and decoder. Because the output of the final layer of the encoder is a compressed representation of sensor data over time, such information is useful for smoothing the estimates.

3.5.2. Motif-guided training

To enhance the training of the network, the attention heads in MGA-Net and MGA-Net+ are trained to detect motifs using motif-occurrence sequences. We prepared three occurrence sequences corresponding to unique, supportive, and boundary motifs for each operation and period. These sequences specify where the motifs occur in a period, and they can therefore be useful for training the attention heads prepared for the operation. Using Adam [37], we train MGA-Net+ through backpropagation by minimizing the loss function calculated based on the occurrence sequences and estimates:

$$E(\theta, \mathbf{o}) = \frac{1}{N_T}\sum_{i=1}^{N_T}\mathcal{L}_c(\theta) + \lambda_2\frac{1}{N_T}\sum_{i=1}^{N_T}\mathcal{L}_a(\theta_a, \mathbf{o}^i) + \lambda_3\frac{1}{N_T}\sum_{i=1}^{N_T}\mathcal{L}_s(\theta), \tag{6}$$

where θ is a set of trainable parameters of MGA-Net+, N_T is the number of training periods, θ_a represents the trainable parameters of the attention heads in θ, $\mathbf{o}^i$ is a set of motif-occurrence sequences during the ith period, and λ_2 and λ_3 are the tradeoff hyperparameters of the three loss functions. In addition, $\mathcal{L}_c(\cdot)$ denotes the cross-entropy loss associated with

Table 2
Overview of the LOGI dataset.

Worker	1	2	3	4
Number of work periods	71	50	48	86
Periods only for training	9	7	12	3
Total duration (s)	2467	3927	5136	6258
Avg duration (s)	34	77	105	72
Worker proficiency	skilled	beginner	mid-level	mid-level
Gender	female	female	female	male
Age	40s	20s	60s	30s
Experience (years)	15	0.5	20	10
Handedness	right	right	left	right

the label prediction for the ith training period. Here, $\mathcal{L}_a(\cdot,\cdot)$ represents the loss involved in attention-guided training for the ith training period, which is formulated as follows:

$$\mathcal{L}_a(\theta_a, \mathbf{o}^i) = \sum_{j=1}^{K*3} \sum_{t=1}^{T} \|\mathbf{a}_t^j - \mathbf{o}_t^{i,j}\|, \tag{7}$$

where $\mathbf{a}_t^j$ is the attention value of the jth attention head at time t, and $\mathbf{o}_t^{i,j}$ is the value of the jth occurrence sequence at time t. Because the sum of the elements in an attention series for a period is 1, we normalize $\mathbf{o}^{i,j}$, which is the jth occurrence sequence, to ensure that the sum of the elements in $\mathbf{o}^{i,j}$ is also 1. Furthermore, MGA-Net+ has three attention heads for each operation (i.e., $K * 3$). Because this loss function denotes the difference between the attention time series computed by the attention head of interest and an occurrence time series of the corresponding motif, we can guide the training of the attention head such that it detects the occurrences of the motif.

Here, $\mathcal{L}_s(\cdot)$ represents the segmentation loss during the ith training period. Segmentation loss is introduced to mitigate the skewed difference among class appearances during the training period [10] and is formulated as follows:

$$\mathcal{L}_s(\theta) = \sum_{t=1}^{T} \sum_{k=1}^{K} -\frac{\mathbf{P}_k}{\mathbf{P}} \ln(f(x)_{k_t}) \tag{8}$$

where $\mathbf{P}_k$ is the number of data points belonging to class k within the ith period, and $\mathbf{P}$ is the total number of data points belonging to any class within the ith period. The combination of these three losses allows us to fine-tune the parameters required to train the segmentation network and the attention mechanism. Thus, to accurately determine the start and end times of each operation, we can identify the locations of the key actions that describe each operation while satisfying the requirements.

4. Evaluation

4.1. Dataset

We evaluated the proposed method using two logistics domain datasets with acceleration data: the LOGI and LARa datasets. The LOGI dataset is a private dataset that was collected from data gathered by four workers at a real logistics center following an instruction manual with sequential operations. Data were collected using a smartwatch (Sony SmartWatch3 SWR50) worn by each worker on their dominant hand, with a sampling rate of 30 Hz. The collected data were manually labeled to generate the ground truth using video recordings. The total duration of the dataset is approximately 5 h with each individual period lasting an average of 87 s. This dataset contains 10 operations (activity classes) to recognize: "Picking", "Replace label", "Assemble box", "Pack in box", "Close box", "Attach label", "Read label", "Put on cart", "Attach address label", "Use pen", and "Unassigned (others)". Table 2 presents an overview of this dataset. Note that sensor data from certain periods are used only for training because of issues related to a data collection failure or labeling. Fig. 8 shows the number of instances contained by each class based on the worker. We can observe how classes such as "Picking" and "Put on cart" contain a much smaller number of instances on all workers. However, each worker has other classes with an extremely low presence, for example, "Use pen" and "Attach label" for Worker 4, it is the short-duration operations that will be the focus of our analysis of the results. In this study, short-duration operations are defined as those classes whose total number of instances in each training period correspond to less than 10% of the total instances for all classes.

The **L**ogistic **A**ctivity **R**ecognition Cha**l**lenge (LARa) dataset [38] is a publicly available dataset that contains accelerometer data from 14 subjects conducting product picking and packaging on 3 different simulated scenarios. The data were captured using MMRL IMUs of MbientLab at an approximate sampling rate of 100 Hz. For the evaluation, we used the accelerometer data corresponding to the dominant wrist from six of the eight workers who conducted scenarios 2 and 3 for a total of 168 working periods or 5.6 h of data, with each period lasting an average of 120 s. We selected these

Table 3
Overview of the LARa dataset.

Worker	7	8	9	10	11	12	13	14
Number of work periods	27	27	27	25	12	20	28	28
Total duration (s)	3187	2843	3012	1998	1423	1824	3163	2992
Avg duration (s)	118	118	115	111	118	107	117	115
Gender	male	female	male	male	female	female	female	male
Age	23	51	35	49	47	23	25	54
Handedness	right	right	right	right	right	right	right	right

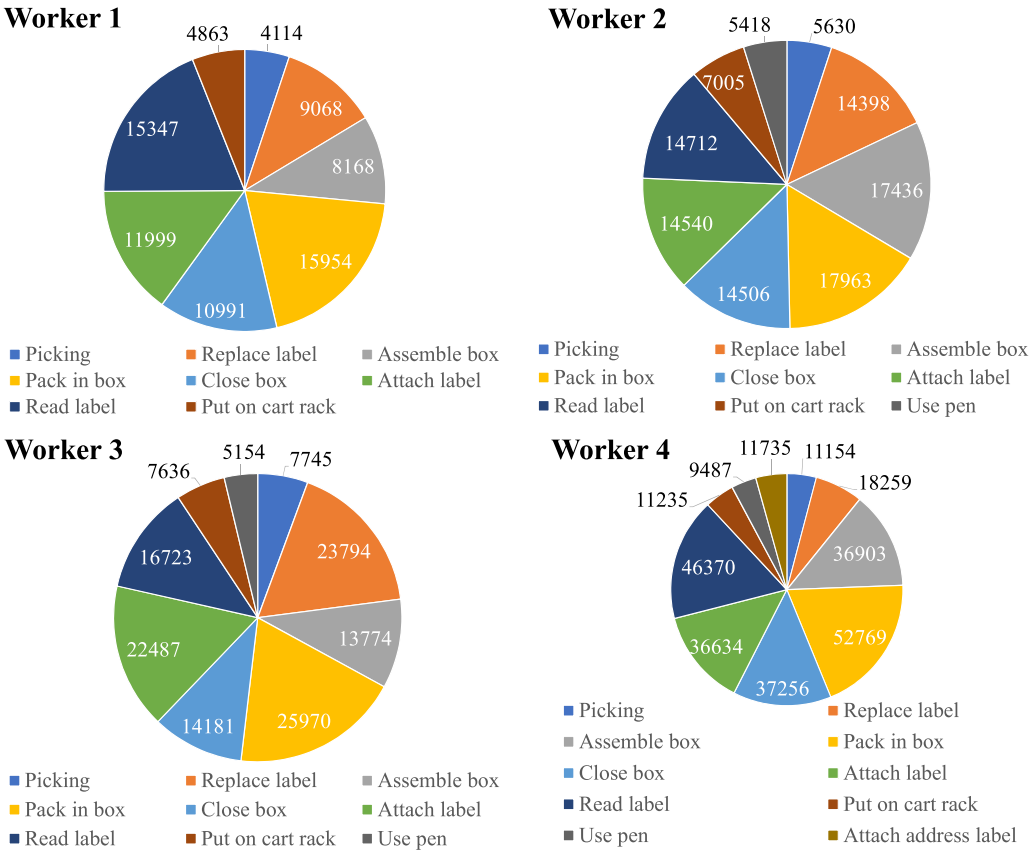


Fig. 8. LOGI dataset class distribution by worker.

scenarios given the similarity of the operations conducted with those of the LOGI dataset, such as "Scanning a code/label" or "Seal/close box". Similar to the LOGI dataset, these scenarios were also defined as sequential operations. This dataset contains six operations (activity classes) to recognize: "Standing", "Walking", "Cart", "Handling (upward)", "Handling (centered)", and "Handling (downward)". Table 3 presents an overview of this dataset. Fig. 9 shows the number of instances based on the operation for the LARa dataset. We can see here that each worker has different under-represented activities, with "Handling (downward)" being the most common. Note that workers 11 and 12 were excluded from the evaluation because of their small number of working periods derived from excessive noise and faulty sensor readings.

4.2. Evaluation methodology

We used a leave-one-period-out cross-validation (i.e., worker-dependent models) and leave-one-worker-out cross-validation for the evaluation. All methods were evaluated using the weighted average F1-score based on an estimate for each sample. To evaluate the effectiveness of the proposed method, we compared it with both window- and sample-based approaches.

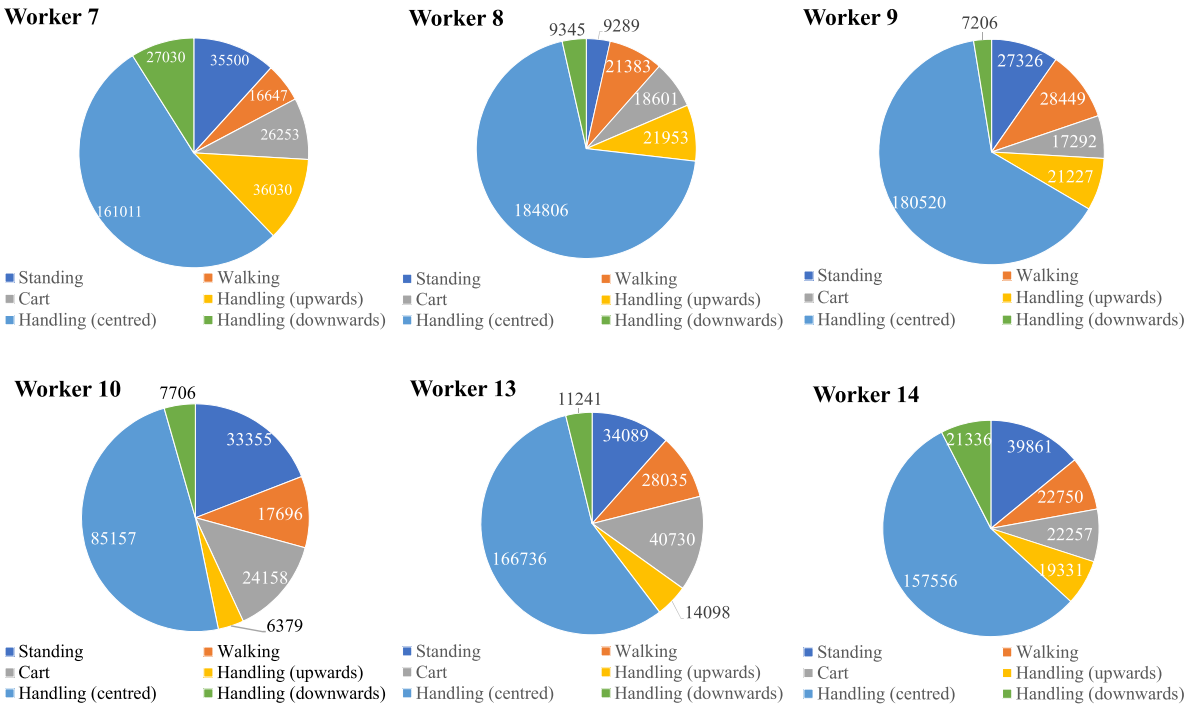


Fig. 9. LARa dataset class distribution by worker.

Table 4
Experimental parameters used for evaluation.

Parameter	Value	Parameter	Value
PAA window size	10	t_{su}	2 s
# symbols	10	f	30
l_m	[2,3,5,7,10,15,20]	λ_2	125
th_s	0.9	λ_3	1

4.2.1. Window-based methods

The input of the window-based method is a sensor-data segment in a time window with a size of 30 data points. We used a sliding time window with a stride length of one point. For window-based methods using deep learning, we employed a cross-entropy loss function based on the Adam optimizer. The learning rate was 0.01, and the training period was 100 epochs, with a batch size of 128.

- **LSTM**: We used a network derived from [39] LSTM-F composed of four LSTM layers with the sigmoid activation function and a softmax output layer.

- **DeepConvLSTM**: This is a more recent baseline and is considered state-of-the-art for HAR using convolutional LSTM networks. We selected a variation of the model applied in citet Pascanu.

- **Conv-IMU**: This method [7] uses parallel branches for multi-channel multi-sensor data streams.

Although the advantage of using parallel branches dedicated to multiple IMUs is not applied when we use only one IMU in our experiment, we consider Conv-IMU to achieve a state-of-the-art performance owing to its ability to reduce the temporal variability of the predictions with its asynchronous loss calculation (see [7] for further details). We therefore used this as the baseline for comparison.

- **Random Forest**: The random forest algorithm was applied. A total of 100 trees number were used. The statistical features were extracted from each window. The standard deviation, zero-crossings, and energy were extracted for each axis, along with the root mean square for the three axes based on prior studies [40,41].

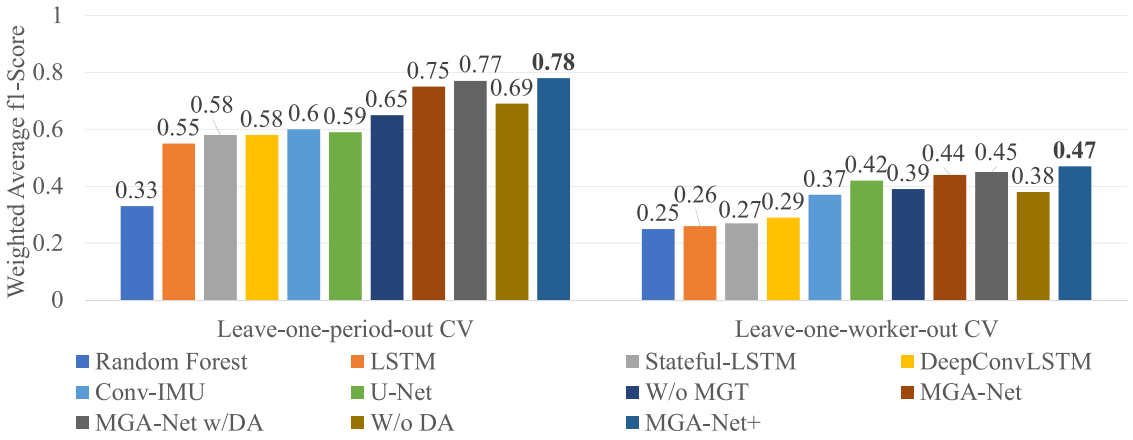


Fig. 10. Average F1-score for all tested methods for LOGI dataset on both evaluation tests.

4.2.2. Sample-based methods

The input of the sample-based method is a sensor data segment corresponding to one work period. The sample-based method outputs a class estimate for each data point. The training period used was of 100 epochs with a batch size of 4. A batch corresponds to a sensor data segment of one period. The Adam optimizer was used, with a learning rate of 0.01.

- **MGA-Net+**: This is the proposed method.

- **Stateful-LSTM**: This is the variant of the LSTM method. We used the same network structure as applied with the LSTM method with stateful training enabled.

- **U-Net**: This is the base model of our method and is based on a neural network for time series segmentation [10]. The network consists of three encoding blocks and three decoding blocks. For each encoder and decoder block, we used two 1D convolutional layers with a kernel size of 3 and a max-pooling/upsampling layer with a kernel size of 2.

- **MGA-Net**: Reference method proposed in [9]. The original Motif-Guided Attention Network model without data augmentation or bi-directional GRU blocks for smooth short-operation prediction.

- **W/o MGT**: This is a variant of the reference method introduced in [9] without motif-guided training. This method is regarded as a fusion of state-of-the-art segmentation [10] and attention [42] HAR networks. The network architecture and parameters are identical to those of MGA-Net.

- **MGA-Net w/DA**: This is a variant of the reference method MGA-Net introduced in [9] with the same data augmentation but without the recurrent blocks introduced in MGA-Net+.

- **W/o DA**: This is a variant of the proposed method without data augmentation. This method applies the model used in MGA-Net+ to only the original training data.

Table 4 lists the experimental parameters.

4.3. Results

4.3.1. Leave-one-period-out CV

The left part of Fig. 10 shows the mean F1-scores over all four workers in the LOGI dataset, showing that MGA-Net+ significantly outperformed the other methods. Similarly, the left side of Fig. 11 shows the mean F1-scores for all four workers in the LARa dataset. MGA-Net+ slightly outperforms the previous methods. Table 5 shows the mean F1-score of all methods for the LOGI dataset. Surprisingly, MGA-Net+ achieved F1-scores of higher than 0.80 for workers 1, 3, and 4, despite the observed data being extremely complex compared to data used in standard HAR studies, and it outperformed the original MGA-Net. Specifically, MGA-Net+ significantly outperformed the other methods for Worker 3. This could have occurred because the duration of the operations performed by the worker was occasionally longer than that of the other workers. Therefore, the effect of the Bi-GRU blocks in MGA-Net+ was high for Worker 3. In addition, the F1-scores of short-duration operations for MGA-Net+ were high, as described below.

As shown in Table 5, the F1-scores of the state-of-the-art methods (DeepConvLSTM, Conv-IMU, U-Net, and W/o MGT) were lower than 0.7 in many cases. The mean F1-score for the LSTM method was 0.55, indicating the difficulty of this task. The mean F1-score for the Stateful-LSTM method was slightly better than that of LSTM, suggesting that, compared with LSTM, Stateful-LSTM can capture the long-term dependency in the data because its input is longer than that of LSTM. The mean F1-score for the Random Forest method was 0.33. As expected, the traditional machine learning method lacks the

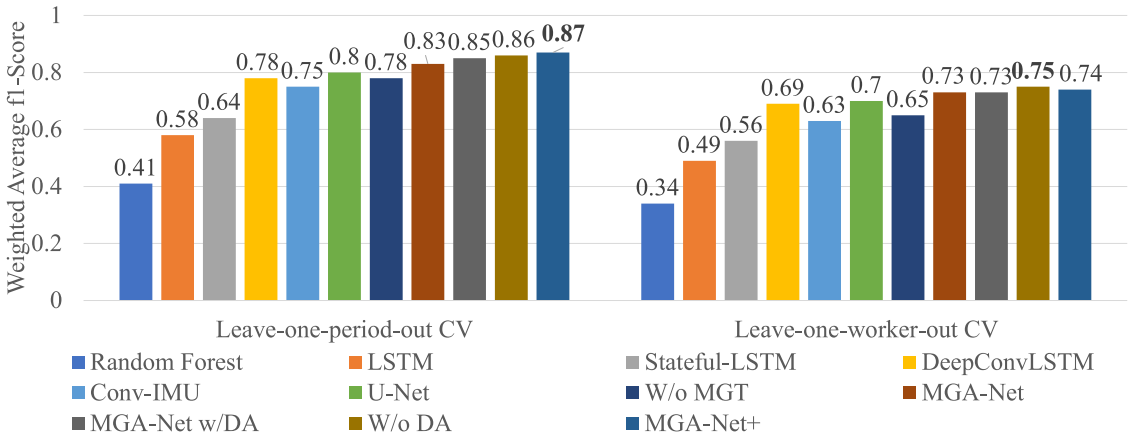


Fig. 11. Average F1-score for all tested methods for LARa dataset on both evaluation tests.

Table 5
Weighted average F1-score for all tested methods and all workers for the LOGI dataset. These are the results of the leave-one-period-out cross validation.

Worker	1	2	3	4	AVG
Random Forest	0.33	0.31	0.32	0.36	0.33
LSTM	0.57	0.51	0.53	0.61	0.55
Stateful-LSTM	0.60	0.54	0.57	0.63	0.58
DeepConvLSTM	0.58	0.55	0.53	0.66	0.58
Conv-IMU	0.56	0.46	0.65	0.69	0.60
U-Net	0.67	0.47	0.60	0.65	0.59
W/o MGT	0.73	0.54	0.63	0.69	0.65
MGA-Net	0.83	**0.62**	0.74	0.81	0.75
MGA-Net w/DA	**0.85**	0.61	0.79	**0.84**	0.77
W/o DA	0.75	0.53	0.76	0.72	0.69
MGA-Net+	**0.85**	0.61	**0.85**	0.81	**0.78**

Table 6
Weighted average F1-score for selected methods and workers from LARa dataset. These are the results of the leave-one-period-out cross validation.

Worker	7	8	9	10	13	14	AVG
Random Forest	0.42	0.38	0.40	0.42	0.40	0.41	0.41
LSTM	0.50	0.62	0.57	0.63	0.59	0.56	0.58
Stateful-LSTM	0.56	0.67	0.62	0.69	0.65	0.62	0.64
DeepConvLSTM	0.79	0.77	0.74	0.76	0.79	0.81	0.78
Conv-IMU	0.77	0.72	0.76	0.75	0.79	0.74	0.75
U-Net	0.80	0.81	0.80	0.81	0.81	0.80	0.80
W/o MGT	0.78	0.77	0.80	0.78	0.79	0.79	0.78
MGA-Net	0.84	0.82	0.83	0.85	0.85	**0.82**	0.83
MGA-Net w/DA	0.87	0.79	0.86	**0.92**	**0.87**	0.81	0.85
W/o DA	0.9	0.82	0.86	0.91	**0.87**	0.81	0.86
MGA-Net+	**0.92**	**0.86**	**0.87**	0.89	**0.87**	**0.82**	**0.87**

ability to properly recognize this type of complex data. As indicated based on the results, attention-based methods (W/o MGT, MGA-Net, MGA-Net w/DA, W/o DA, and MGA-Net+) outperform the simpler methods. Specifically, motif-guided training compensates for the scarcity of training data. The mean F1-score of the workers for MGA-Net+ was approximately 20% higher than the mean F1-scores of the state-of-the-art methods. In addition, MGA-Net+ outperformed W/o MGT and W/o DA by approximately 10%, indicating the significant contribution of motif-guided training and data augmentation. Interestingly, the performance of the W/o DA was poorer than that of the original MGA-Net. This could be because the number of parameters of MGA-Net+ increased when introducing the Bi-GRU blocks. The data augmentation approach can alleviate the issue of model complexity.

Table 6 shows the F1-scores for the leave-one-period-out cross validation for the LARa dataset. When comparing the results with those of LOGI, the performance of MGA-Net+ appears to be more stable across all workers. The MGA-Net+ and W/o DA models were able to achieve a consistent improvement over the state-of-the-art approach for all workers, except for Worker 14. Interestingly, with LARa, we see the opposite effect when analyzing the impact of the two new method contributions (data augmentation and GRU blocks). MGA-Net w/DA shows an overall worse F1-score than the

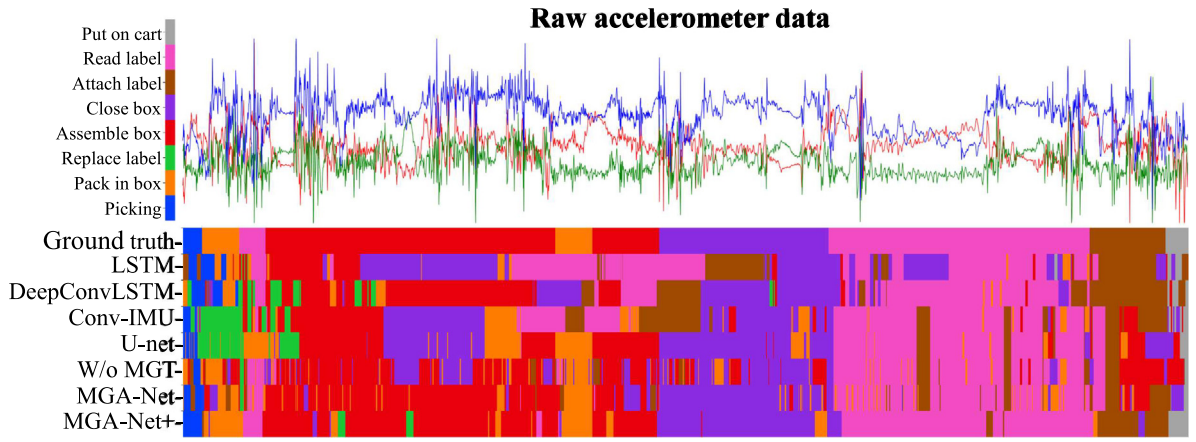


Fig. 12. Ground truth and predictions based on the method applied for an example working period from Worker 4 (LOGI dataset). The horizontal axis shows the time.

W/o DA model, particularly for workers 7 and 8. This means that GRU blocks have a higher impact on the results than data augmentation. Such a difference can be explained by the class distribution shown in Fig. 9, where most of the classes from Scenarios 2 and 3 belong to "Handling (centered)", which is usually presented as a single continuous label for more than 30% of the total period duration. Table 6 also shows that methods with no convolutional layers have a much worse performance with an average F1-score of 0.54, as stated based on the results for Random Forest, LSTM, and Stateful-LSTM. However, the methods with both recurrent and convolutional strategies (DeepConvLSTM, W/o DA, and MGA-Net+) outperformed the pure convolutional methods, with the hybrid methods averaging an F1-score of 0.83 and convolutional methods averaging a score of 0.8.

Fig. 12 shows the ground-truth labels and predictions using the methods applied for the work period of Worker 4 from the LOGI dataset. As shown in Fig. 12, the U-Net results contain many false detections with a long duration owing to the overall similarity in sensor data between different operations. The results of Conv-IMU, LSTM, and DeepConvLSTM also contain many false detections with a long duration. Even when we used an attention-based neural network for segmentation (i.e., W/o MGT), we were unable to prevent false detections with a short duration, which may occur because, although self-attention can help with finding the key actions, the number of data required for a good performance is much higher than the number of data in the LOGI dataset. MGA-Net and MGA-Net+ capture the overall trend of the operational sequence. However, the results of MGA-Net still contain false detections with a short duration, particularly in the latter half of the results. With the addition of Bi-GRU blocks, MGA-Net+ can output smoothed estimates, as shown in Fig. 12.

Fig. 13 shows the F1-scores for short-duration operations for each worker in a leave-one-period-out cross-validation. We can see that compared to MGA-Net, both MGA-Net with data augmentation (**MGA-Net w/DA**) and MGA-Net+ have a significant improvement with most of these operations. Particularly for the "Put on cart" and "Use pen" operations, we observe a 5%–8% increase in the F1-score. As mentioned in Section 4.1, each worker has a different class distribution, which means that each new copy of a period generated during the data augmentation impacts the outcome of the training for short-duration operations. We can observe this effect when we compare the results of MGA-Net+ for workers 2 and 3. Worker 2 shows a small improvement for short-duration operations, whereas the overall average F1-score across all operations seems to be lower than that achieved through MGA-Net. By contrast, Worker 3 shows a larger improvement in short-duration operations, as well as an overall average F1-score higher than that of all other methods. Another reason may be the effect of the introduced Bi-GRU blocks on the recognition of longer operations. Worker 3 shows the highest performance when testing the MGA-Net+ model without a data augmentation or "W/o DA" because Worker 3 is an elderly person with mid-level knowledge of the process and whose average period duration is considerably longer than that of any of the other workers, which translates into longer operations that are better suited for the GRUs introduced into the network architecture even without a data augmentation.

4.3.2. Leave-one-worker-out CV

The right side of Fig. 10 shows the mean F1-scores for all four workers for the LOGI dataset. By contrast, the right side of Fig. 11 shows the mean F1-scores for all workers from the LARa dataset. In addition, Table 7 presents the results for each of the four workers in the LOGI dataset. Table 8 shows the results of the method used for the six workers in the LARa dataset. In both tables, we can see that the F1-scores for leave-one-worker-out cross-validation were poorer than those for leave-one-period-out cross-validation because of the high degree of freedom for packaging work, resulting in large differences in the sensor data between workers. For example, the method of assembling shipping boxes greatly depends on the worker preferences. We can observe that the performance of W/o MGT is poor compared to that of U-Net. By contrast, the performance of MGA-Net+ was notably higher than that of U-Net. This indicates that capturing the key

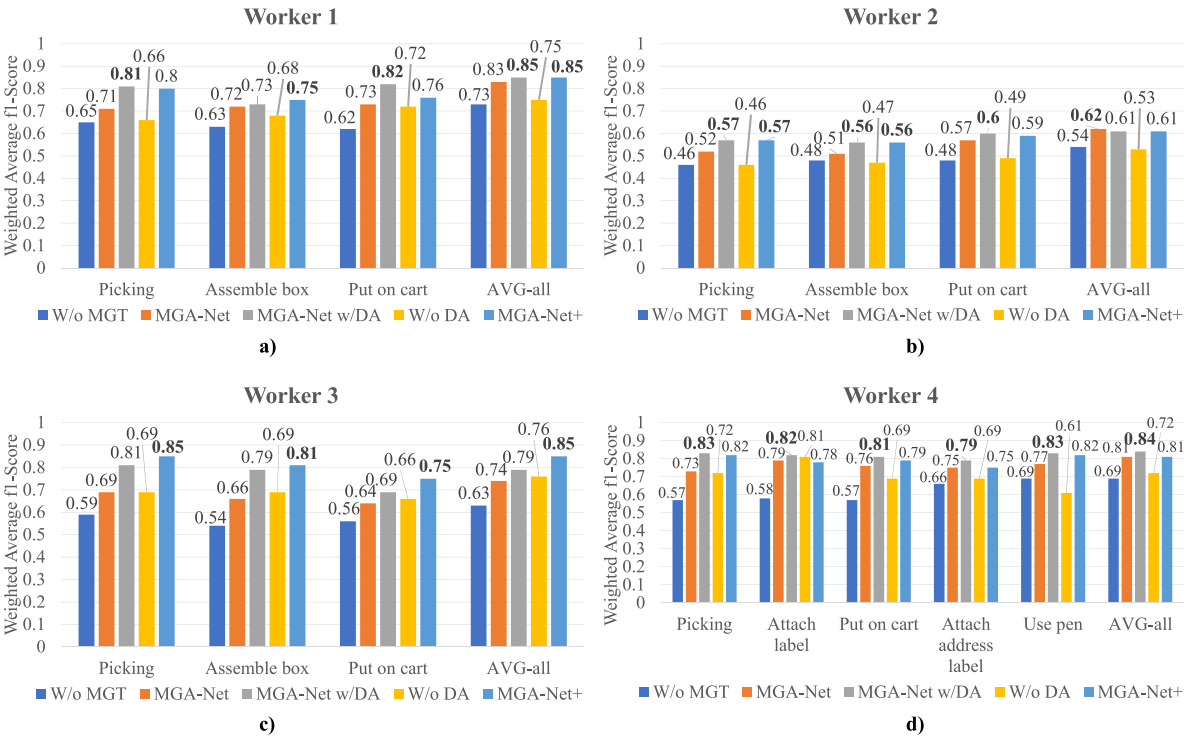


Fig. 13. F1-scores of short-duration operations for all workers and methods. This is the result of the leave-one-period-out cross validation from the LOGI dataset.

Table 7
Weighted average F1-score for all methods and workers from LOGI dataset. The results of leave-one-worker-out cross validation.

Worker	1	2	3	4	AVG
Random Forest	0.25	0.22	0.25	0.28	0.25
LSTM	0.25	0.12	0.32	0.36	0.26
Stateful-LSTM	0.23	0.17	0.3	0.37	0.27
DeepConvLSTM	0.27	0.16	0.32	0.39	0.29
Conv-IMU	0.33	0.34	0.39	0.41	0.37
U-Net	0.51	0.36	0.40	0.42	0.42
W/o MGT	0.45	0.31	0.43	0.35	0.39
MGA-Net	0.51	0.39	0.46	0.40	0.44
MGA-Net w/DA	0.49	0.41	0.46	0.47	0.45
W/o DA	0.43	0.35	0.34	0.39	0.38
MGA-Net+	**0.52**	**0.43**	**0.47**	**0.53**	**0.47**

Table 8
Weighted average F1-score for selected methods and workers from LARa dataset. The results of leave-one-worker-out cross validation.

Worker	7	8	9	10	13	14	AVG
Random Forest	0.35	0.33	0.36	0.30	0.33	0.33	0.34
LSTM	0.43	0.56	0.48	0.44	0.51	0.49	0.49
Stateful-LSTM	0.55	0.63	0.48	0.61	0.47	0.64	0.56
DeepConvLSTM	0.67	0.68	0.67	0.68	0.71	0.73	0.69
Conv-IMU	0.60	0.63	0.67	0.61	0.68	0.62	0.63
U-Net	0.72	0.72	0.68	0.70	0.69	0.68	0.70
W/o MGT	0.69	0.68	0.65	0.60	0.66	0.65	0.65
MGA-Net	0.75	0.75	0.73	0.67	0.77	0.73	0.73
MGA-Net w/DA	**0.78**	**0.76**	0.73	0.66	0.76	0.71	0.73
W/o DA	0.75	**0.76**	**0.75**	0.7	**0.78**	**0.75**	**0.75**
MGA-Net+	0.76	**0.76**	0.71	**0.72**	0.76	0.73	0.74

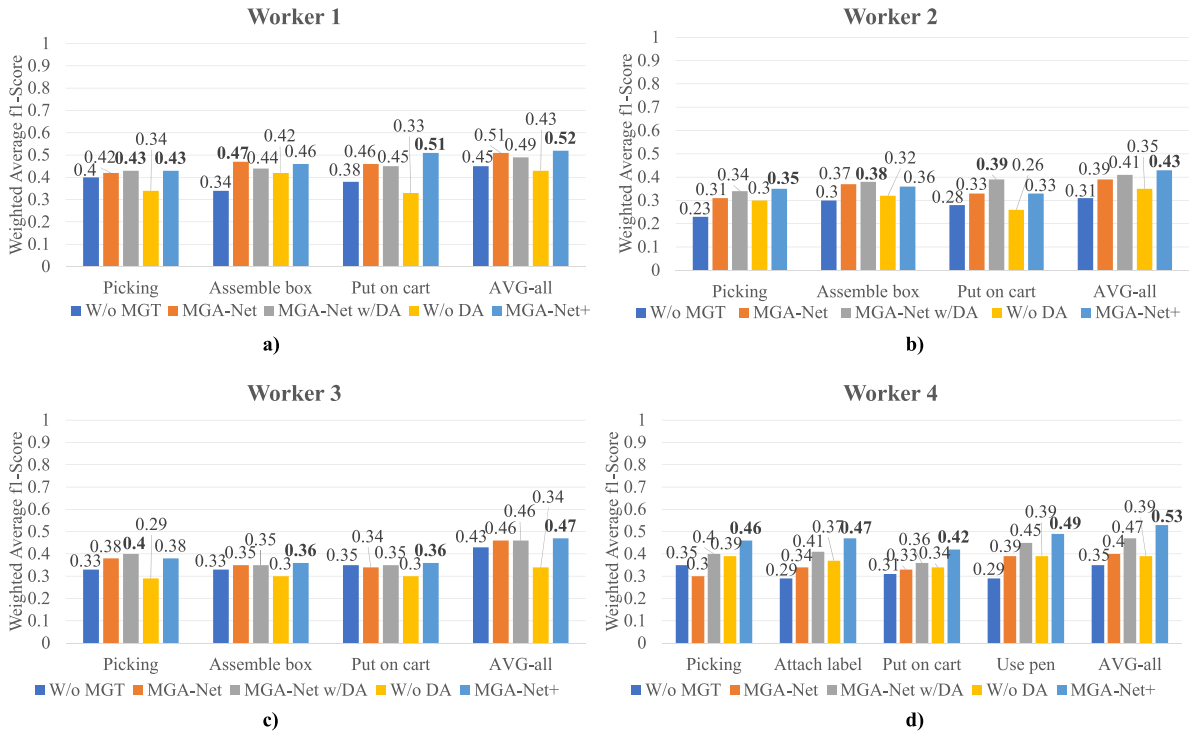


Fig. 14. Weighted average F1-score of short-duration operations of all workers and methods. This is the result of the leave-one-worker-out cross validation from the LOGI dataset.

actions commonly found in different workers using a self-attention-based method (i.e., W/o MGT) is difficult. However, motif-guided training appeared to facilitate the detection of key actions commonly found in training workers. In addition, for the results of the LOGI dataset, the performance of W/o DA was poorer than the original MGA-Net and U-Net. This is surprising given that for the LARa dataset, W/o DA outperformed all other methods. As mentioned above, for the LARa dataset, the impact of the GRU blocks seems to significantly improve the original MGA-Net model even with limited training data.

Focusing on the results from the LOGI dataset shown in Table 7, we can see how the performance of MGA-Net+ for Worker 4 was much higher than that of the other methods, particularly when considering the improvement compared to MGA-Net among all four workers. As can be seen in Fig. 14, Worker 4 is the only worker that shows a significant increase in the prediction results for short-duration operations when combining the data augmentation and Bi-GRU blocks. The results of MGA-Net w/DA and MGA-Net for short-duration operations indicate that data augmentation contributed to the approximately 5% improvement in the F1-score. Because MGA-Net+ outperformed MGA-Net by approximately 10%, we can say that the impact of the combination of data augmentation and Bi-GRU blocks is significant. In Table 8, we can see that only workers 7, 8, and 10 have a higher performance for MGA-Net+ than for MGA-Net, with Worker 10 showing an increase of approximately 17%.

When looking more deeply at the differing results from workers in the LARa dataset (i.e., W/o DA versus MGA-Net w/DA), we were unable to find relevant worker characteristics that can explain such differences. Unlike the LOGI dataset, where worker experience shows a clear impact on the results for both the leave-one-period-out and leave-one-worker-out experiments, the LARa dataset does not include information on worker skills or experience, and other available characteristics such as gender, age, and body proportions show no significant impact on the results.

4.4. Discussion

4.4.1. Contributions of motifs

Here, we investigate the contributions of these three types of motifs. The following methods were prepared:

- **MGA-Net+ (U)**: A variant of the proposed MGA-Net+ that uses only unique motifs.
- **MGA-Net+ (U+S)**: A variant of the proposed method that does not use boundary motifs.
- **MGA-Net+ (U+B)**: A variant of the proposed method that does not use supportive motifs.
- **MGA-Net+ (B)**: A variant of the proposed method that uses only boundary motifs.

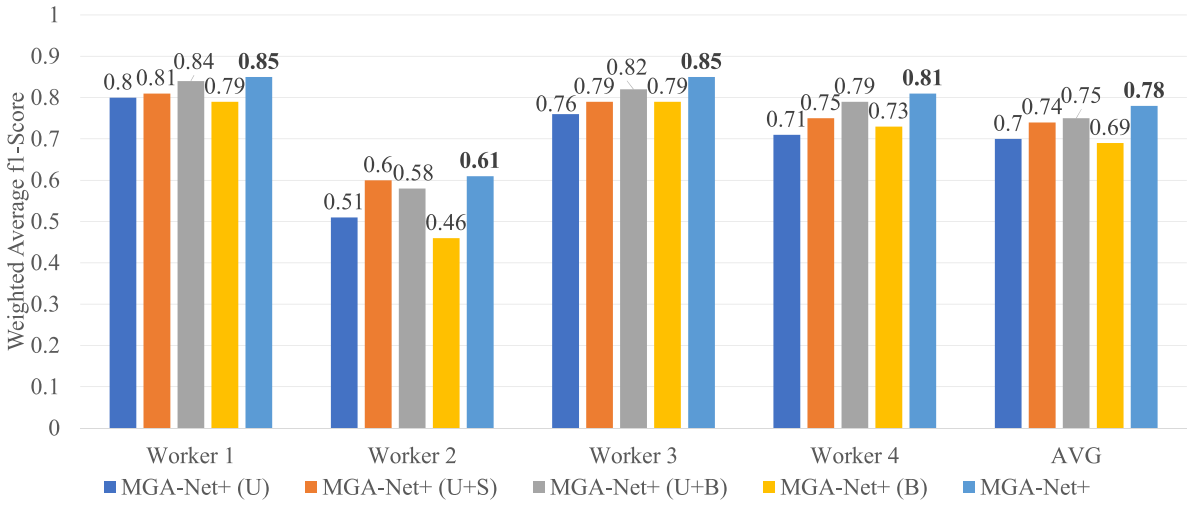


Fig. 15. Contributions of motifs for LOGI dataset. We used 10 randomly selected test periods for this evaluation. The remaining periods from the same worker have been used for training.

Fig. 15 shows the mean F1-scores of the methods and MGA-Net+(U+S+B) for the LOGI dataset. The performance of MGA-Net+ (B) is almost identical to that of MGA-Net+ (U). Because a unique motif occurs almost exclusively in an operation of interest, it is useful to identify a segment corresponding to the operation. Because the identification of the starting and ending times, that is, the boundary, plays a key role in a time series segmentation, boundary motifs also contribute to an accurate segmentation. In the results for Worker 2, the mean F1-score of MGA-Net+ (B) was poorer than that of MGA-Net+ (U). This can be caused by the inexperience of Worker 2, who frequently conducts the same operation multiple times during a single period, rendering the recognition of a single boundary motif meaningless. Focusing on MGA-Net+ (U+B) and MGA-Net+ (U+S), MGA-Net+ (U+S) slightly outperformed MGA-Net+ (U+B), indicating that the contribution of the supportive motifs was greater than that of the boundary motifs when used in combination with the unique motifs. Supportive motifs, given their relation with unique motifs, permit us to capture the temporal structure of an action (i.e., the need to grab a scanner before scanning a label), which proves effective when trying to predict the duration of an operation.

4.4.2. Analysis of motif identification

Here, we analyze the results of the motif identification method. As mentioned in Section 3.3, because the process for selecting a motif varies based on the operation and motif type, various motions with different lengths can be identified as relevant motifs. Fig. 16 shows an example of the output from the motif identification method for a "Close box" operation (motif occurrence sequences prior to thresholding). We can observe that both the unique and boundary motifs were able to identify specific atomic actions that can be attributed to the "Close box" operation. For example, the pressing of the cushion materials before the closing of the sides of the box can be the start of the operation during every iteration and is a unique action that can only be present in a "Close box" operation. In the case of actions found through unique and supportive motifs, we can understand the relative importance to the operation given that only two operations in the sequence use tape; however, the simplicity of the motion makes it difficult to find a perfect match and opens the possibility for incorrect appearances for the same or different operations.

In Fig. 16, a motif occurrence sequence of the boundary motif shows a similarity score of "1.0" in its region, which indicates a perfect match to its original seed, whereas both the unique and supportive motifs have "0.9" matches within the operation. All three motifs have no matches in the adjacent operations; however, we note that for the selected occurrence sequence of the supportive motif, there are matches with "0.9" and "1.0" similarity values in two other operations ("Assemble box" and "Replace label") not visible in the figure. The appearance of high similarity values for other operations/actions for a supportive motif can be attributed to the selection method, whereas unique and boundary motifs are selected, giving a high priority to the relative number of appearances outside the target operation. By contrast, the supportive motif mostly considers the dependence of time and appearance on a unique motif.

4.4.3. Contributions of attention

Here, we qualitatively analyze the attention value output using the attention head of MGA-Net+. To acquire the values for such an analysis, we first obtained a set of attention values in each label of an operation for which the attention head is responsible. Subsequently, we computed the average and maximum attention values for the set of attention values. Finally, we computed their average (i.e., the average of the mean attention values and the average of the maximum attention values) over all instances of the operation of interest. Moreover, we computed the average and maximum values outside

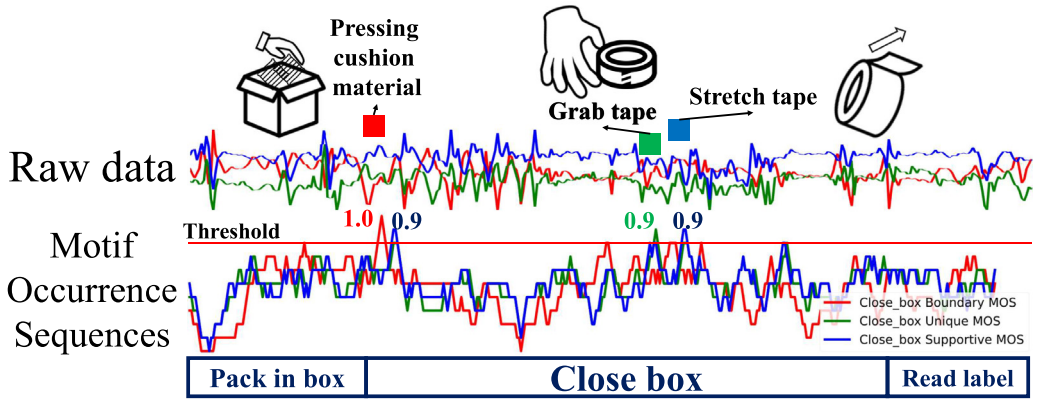


Fig. 16. Example of motif occurrence sequences (MOSs) for "Close box" operation for Worker 1. Relevant atomic actions found by each motif type within an operation are highlighted.

Table 9
Analysis of attention values in MGA-Net+ for the LOGI dataset.

	Unique motif	Supportive motif	Boundary motif
Avg. in Op. of interest	0.398×10^{-3}	0.421×10^{-3}	$\mathbf{0.533} \times 10^{-3}$
Max. in Op. of interest	$\mathbf{13.59} \times 10^{-3}$	$\mathbf{3.17} \times 10^{-3}$	$\mathbf{5.36} \times 10^{-3}$
Avg. outside Op. of interest	0.137×10^{-3}	0.196×10^{-3}	0.243×10^{-3}
Max. outside Op. of interest	2.76×10^{-3}	0.815×10^{-3}	0.639×10^{-3}

Table 10
Analysis of attention values in MGA-Net+ for the LARa dataset.

	Unique motif	Supportive motif	Boundary motif
Avg. in Op. of interest	$\mathbf{0.658} \times 10^{-3}$	0.395×10^{-3}	0.337×10^{-3}
Max. in Op. of interest	$\mathbf{3.22} \times 10^{-3}$	$\mathbf{1.84} \times 10^{-3}$	$\mathbf{3.51} \times 10^{-3}$
Avg. outside Op. of interest	0.062×10^{-3}	0.111×10^{-3}	0.197×10^{-3}
Max. outside Op. of interest	0.854×10^{-3}	0.129×10^{-3}	0.476×10^{-3}

the labels of the operation of interest. Table 9 shows the results acquired from this analysis for the LOGI dataset. The maximum attention value in the labels of the operation of interest was approximately 10-times higher than that outside the labels, indicating that the trained attention heads can focus on operations for which they are responsible. Specifically, the maximum attention value for the boundary motifs in the labels of the operation of interest was approximately 10-times higher than that outside the labels. This may be because the attention heads for the boundary motifs focus only on short-duration segments around the operational boundaries. Table 10 shows the results acquired through this analysis for the LARa dataset. LARa is a unique motif that shows a considerable increase in the average attention value inside the operation of interest, being more than 10-times that outside the operation of interest. This indicates that it is easier to find key actions that are representative of the label in question. However, if we observe the results for the maximum attention value for the unique and supportive motifs compared to that of the LOGI dataset, the value is considerably smaller. A reason for this can be explained by the attention heads for these motifs finding multiple key actions inside the same operation,

4.4.4. Amount of training data

Fig. 17 shows the transitions in the mean F1-scores of the methods when the number of training periods was randomly reduced. As shown in this figure, MGA-Net and MGA-Net+ outperformed the other methods even when the number of training data was reduced. However, because these deep learning methods are data-intensive, the F1-scores significantly decreased when limited training data were used. The results also indicate that the performance improvement by MGA-Net+ was limited compared to that of MGA-Net when the number of training data was small. This could be because the variation in training data generated by data augmentation was also limited when the number of original training data was small.

4.4.5. Overlapping of operations

In Fig. 12, we can see that the ground truth labels used to test the effectiveness of all methods have defined boundaries. Given the nature of human work, particularly in logistics settings, such as packaging and product picking, boundaries between operations are difficult to define. A worker may reach the barcode scanner before finishing attaching the tape

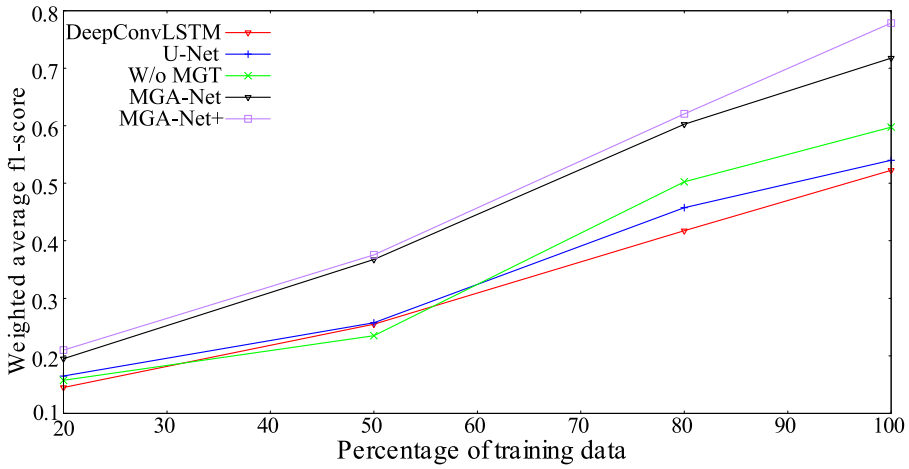


Fig. 17. Transitions in mean F1-scores of the LOGI dataset for each method when we change the number of training data. These results are the averages for the four workers. We used 10 randomly selected testing periods for this evaluation. The remaining periods from the same worker were used as a pool of training periods. We sampled periods from the pool and used them as training data. We changed the percentage of periods sampled from the pool. Note that only sampled training periods that meet the criteria were augmented.

to the box. We observed that such cases occur sporadically and predominantly on experienced workers who feel more familiar with the process. Because the overlapping period is small relative to the length of the operations, for the particular case of this study, we decided to avoid overlapping labels by defining a cutoff point between labels as a particular action that marks the start of the following operation (i.e., touching the scanner). However, even when a time series contains multiple overlapping labels, we believe that we can easily deal with multi-label data by modifying the final layer of the model such that it outputs multiple labels for each data point.

5. Conclusions and future work

This study presented a new method for recognizing complex packaging operations using a body-worn acceleration sensor. We focused on the characteristic actions (motions) that occur in packaging tasks and proposed the use of an attention-based neural network to focus on these characteristic actions when recognizing the data. In addition, the network was designed to capture the temporal regularity of the worker operations and precisely recognize short-duration operations. The proposed method was evaluated on sensor data collected at an actual logistics center, and it significantly outperformed other state-of-the-art methods used in the HAR domain. Although our method can benefit from motifs and focus attention on key actions for individual subjects, the collection and labeling of sufficient data for every type of worker in a logistics center are unfeasible. Providing a method to transfer such abilities across different types of subjects, different sensor settings, and different tasks would further build on the current research results. As a different study branch, the development of a quality assurance tool that can be applied to real-time operation recognition would greatly benefit logistics centers and global supply chains. We highlight these two main avenues as having the potential to improve upon current research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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